

Team name: Avengers

Project name: Sudo

Members:

Dmytro Senyk (Product Owner) - G00439607

Bogdan Tabalae (DevOps) - G00436707

Kostiantyn Panasenko (Quality Tester) - G00436001

Tomasz Touma (Scrum Master) - G00439630

Module name: Project Management

Assignment: Group Project Sub-assignment 05

Lecturer: Joseph Corr

Date: 15-April-2026

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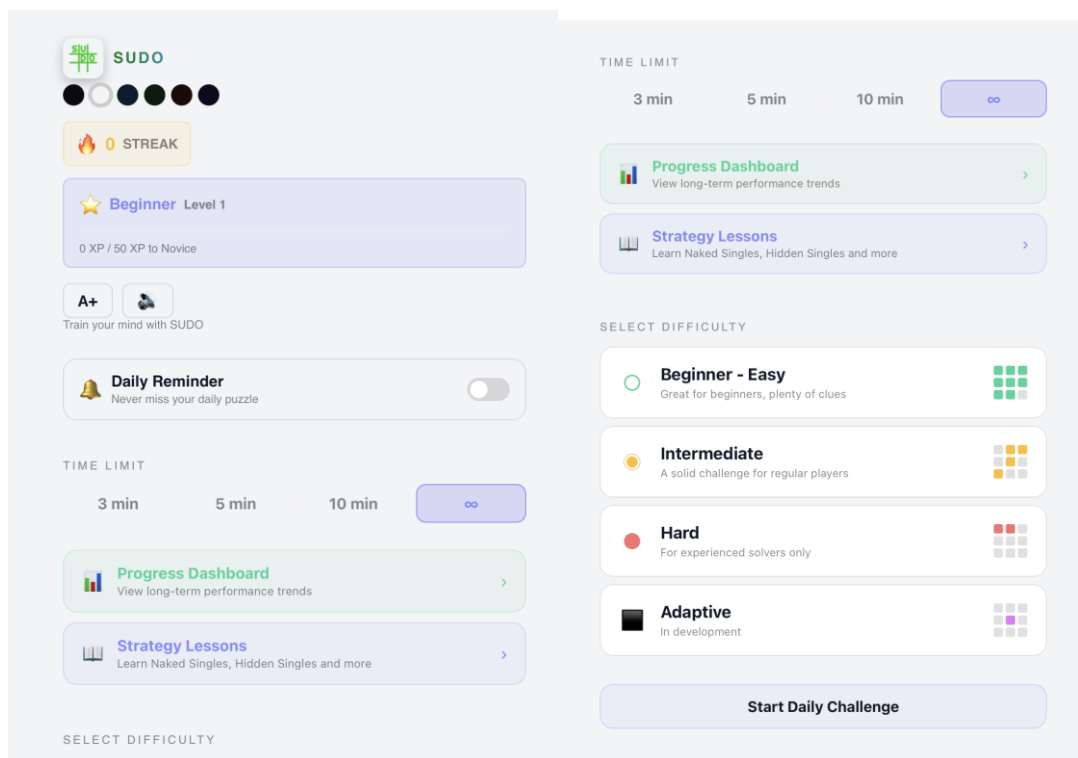
Elevator pitch

Sudo is a personalized brain-training app that dynamically adapts puzzle difficulty to each user's unique skill level.

Users are consistently challenged but never frustrated, driving measurable improvement in logical reasoning, pattern recognition and cognitive agility.

Think of it as a personal trainer for your brain, it meets you where you are, pushes you just enough and grows with you over time.

Product box



STRATEGIES

1 of 3

✕

① Naked Single

When only one number can go in a cell

Ⓢ Naked Single

Ⓜ Hidden Single

Ⓛ Box / Line Reduction

A naked single occurs when a cell has only one possible candidate. Every other digit already appears in the same row, column, or 3x3 box.

5	3			7				
6			1	9	5			
	9	8					6	
8				6				3
4	2	3	1	7	6	7	5	
7				2				6
	6					2	8	
			4	1	9			5
				8			7	9

STEP 1/5

Look at the highlighted cell (row 5, col 5).

💡 Pro tip:

Scan each empty cell and eliminate the digits already present in its row, column, and box. If only one digit remains, place it!

← Back

Reset

Pause

Beginner - Easy

Mistakes: 0/5

10:04

5	4			2	6			7		
8				3	9	1				2
1								8		
					1			5	9	
6			8	5					7	
			1	9			6	2		
9	8	2		6	4					7
7	6				8			9	2	5
3					2					8

💡 Get a hint (5 remaining)

Mistake History

No mistakes yet.

1

2

3

4

5

6

Updated release plan overview

The project was executed across 4 sprints, with a comprehensive backlog of 43+ user stories allocated strategically to balance workload and deliver incremental value.

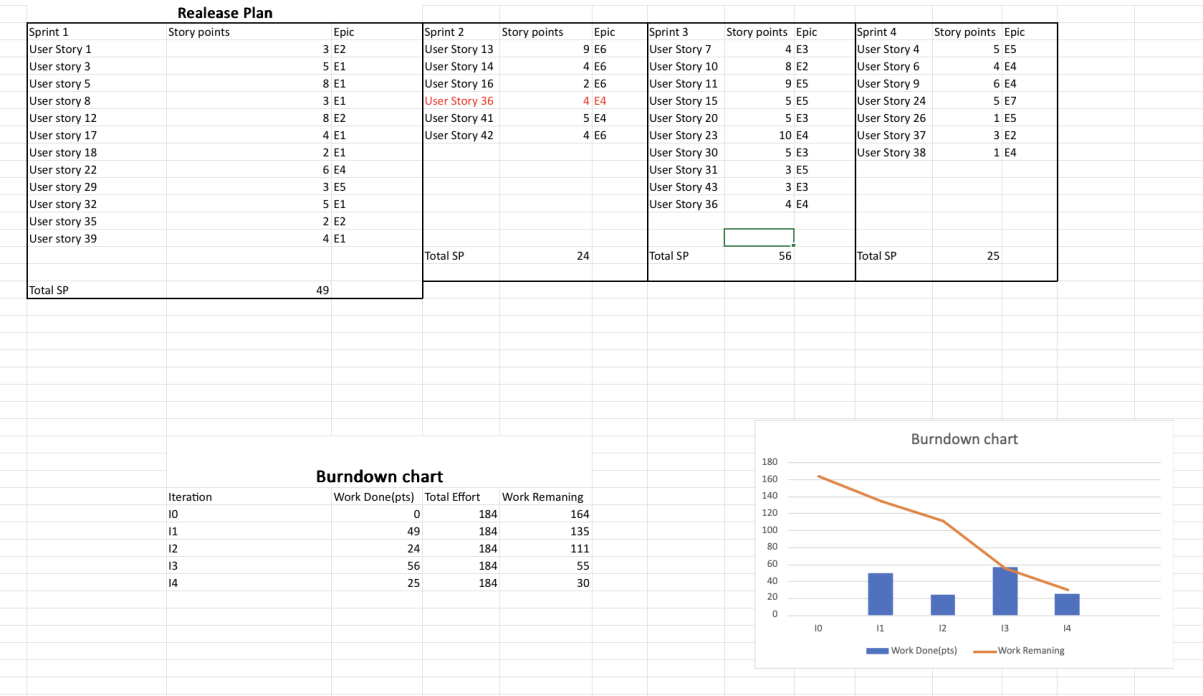
Sprint Allocation

Sprint	Story Points	User Stories	Accumulation
Sprint 1	24 SP	8-10	24
Sprint 2	56 SP	13-14	56
Sprint 3	25 SP	8-10	81
Sprint 4	56 SP	14-15	161

Key Metrics

- Total Story Points: 161 SP across 43+ user stories
- Average Sprint Capacity: 50.25 story points per sprint
- Priority Distribution: Epic (E1-E7) and critical features prioritized in early sprints
- Burndown Chart: Steady progression with consistent story point completion

Release Plan && Burndown Chart



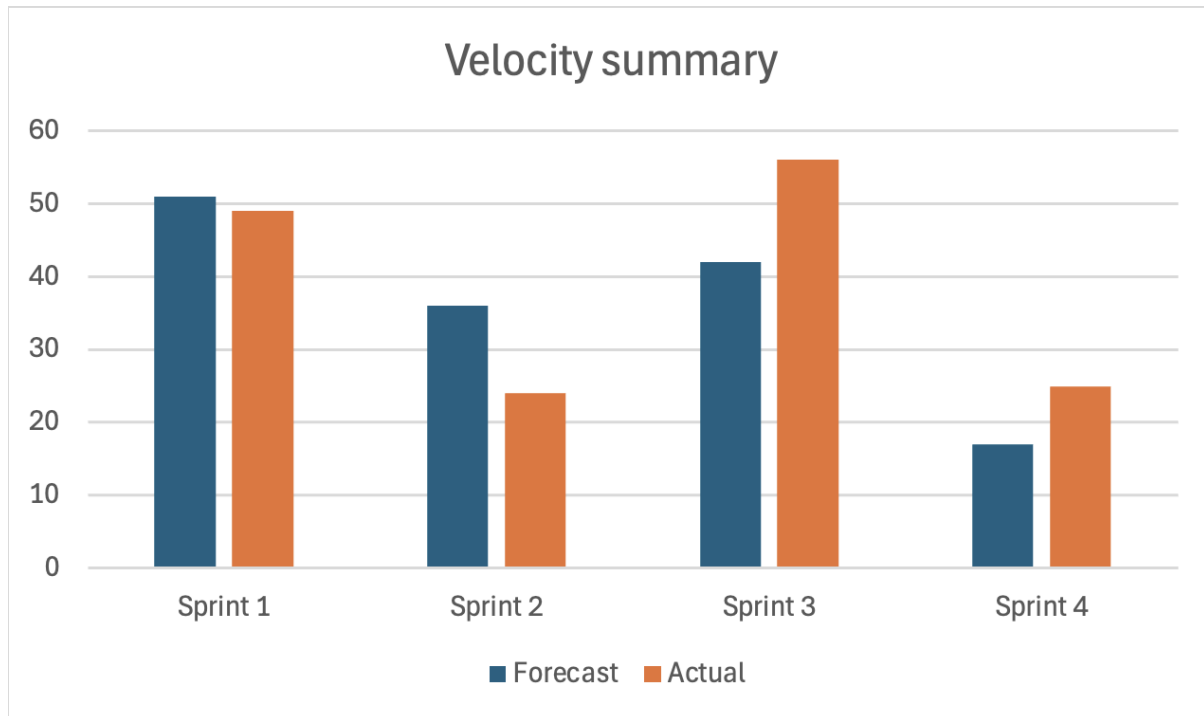
System architecture

Presentation Layer consists of three main React components: GameBoard handles the interactive 9×9 grid, real-time timer, mistake counter and hint interface. HomePage provides the main dashboard with difficulty selection, time limit config, daily reminders and performance stats. Tutorial delivers interactive strategy lessons with step-by-step visualizations.

Game Logic Layer implements core game rules entirely in React. The Game State manages the board array, elapsed time, mistakes, selected cells and game status(won, lost or paused). The Hint System implements two solving strategies — Naked Single and Hidden Single. Progressive hint levels provide cell highlighting, strategy explanation, answer reveal. Validators check for winning conditions and real-time move validation. Difficulty Adaptation gradually reduces available hints over time to encourage player independence.

Data Persistence Layer combines three storage mechanisms. localStorage stores game history, user preferences like theme, font size, sound, daily streaks and notification settings for offline access. The sudokuAPI backend provides puzzle generation, solution validation, and the checkWin() function. Browser APIs including Web Audio API for sound effects, Notifications API for daily reminders.

Velocity summary



Sprint summaries

Sprint 1: Foundation & Core Setup

Sprint Goal: Establish project foundation with basic game mechanics, user authentication, and development workflow

Completed Work:

- Basic Sudoku game board implementation
- Development environment setup
- GitHub repository configuration

Velocity: 24 story points

What Went Well:

- Excellent team communication throughout the sprint
- All assigned stories completed on time
- Jira board provided clear progress tracking
- Team participated actively in discussions

What Could Be Improved:

- GitHub workflow lacked consistency; commits not always linked to stories
- Test cases not written before feature testing
- Meetings were unstructured, causing discussions to run longer than necessary

Actions for Next Sprint:

- Implement GitHub branching strategy with story numbers in branch names
- Include story number in every commit message for better traceability
- Write test cases before development begins and document in OneNote
- Create structured meeting agendas to improve time efficiency

Sprint 2: Core Features & Adaptive System

Sprint Goal: Deliver hint system, and enhance user experience

Completed User Stories (5 of 6):

- SCRUM-13: Dark mode for better usability at night (9 SP) - Dmytro ✓
- SCRUM-42: Accessible font sizes for improved readability (4 SP) - Dmytro ✓
- SCRUM-14: Customizable themes for user personalization (4 SP) - Tomasz ✓
- SCRUM-16: Sound effects toggle to control distractions (2 SP) - Tomasz ✓
- SCRUM-41: Streak counter to encourage consistency (5 SP) - Kostiantyn ✓

Incomplete:

- SCRUM-36: Performance tracking to compare progress (4 SP) - Bogdan ✗

Velocity: 56 story points (incomplete: 60 SP planned)

Deliverables: Sprint 2 demo video uploaded to GitHub (Sprint_2_demo.mp4)

What Went Well:

- Team communication remained effective
- Most user stories completed successfully
- Updated GitHub workflow followed more consistently
- Jira board maintained good task visibility

What Could Be Improved:

- One team member unavailable due to other commitments, causing one story to remain incomplete
- Lack of flexibility in task redistribution when unexpected issues arise
- Not all commits consistently included story references
- Some test case documentation remained brief

Sprint 3: Advanced Features & Testing Infrastructure

Sprint Goal: Implement advanced game features, improve testing automation and enhance user engagement

Completed User Stories (10/10):

- SCRUM-7: Explanations for hints to understand logic (4 SP) - Tomasz ✓
- SCRUM-10: Beginner-friendly puzzles to build confidence (8 SP) - Dmytro ✓
- SCRUM-11: Time-based achievements for fast solving (9 SP) - Kostiantyn ✓

- SCRUM-15: Strategy lessons for advanced skills (5 SP) - Bogdan ✓
- SCRUM-20: Guided tutorials for basic strategies (5 SP) - Dmytro ✓
- SCRUM-23: Mistake history to identify patterns (10 SP) - Bogdan ✓
- SCRUM-36: Performance comparison with past games (4 SP) - Bogdan ✓
- SCRUM-30: Advanced strategy lessons (naked pairs) (5 SP) - Tomasz ✓
- SCRUM-31: Performance-based rewards (3 SP) - Kostiantyn ✓
- SCRUM-43: Tutorial replay option (3 SP) - Dmytro ✓

Velocity: 56 story points

Deliverables: Sprint 3 demo video uploaded to GitHub (Sprint_3_demo.mp4)

What Went Well:

- Improved team collaboration and communication
- Most team members followed GitHub process correctly
- Branch naming and story number usage was clearer and more consistent
- GitHub Actions automated testing implemented successfully
- All planned stories completed

What Could Be Improved:

- Test case documentation in OneNote was incomplete or inconsistent
- GitHub Actions configuration needed optimization for better test reliability

Actions for Next Sprint:

- Continue improving and optimizing GitHub Actions for proper automated testing
- Expand OneNote documentation for test cases and ensure completeness
- Enforce testing completion before merge requests are approved

Sprint 4: Final Features & Deployment

Sprint Goal: Complete final features, set up deployment infrastructure and prepare for release

Completed User Stories (8/8):

- SCRUM-4: Push notifications for daily puzzles (5 SP) - Dmytro ✓
- SCRUM-26: Daily challenges for returning players (1 SP) - Dmytro ✓
- SCRUM-6: Progress dashboard for long-term tracking (4 SP) - Bogdan ✓
- SCRUM-9: Statistics on accuracy (6 SP) - Bogdan ✓
- SCRUM-24: Optional smart hints (5 SP) - Tomasz ✓
- SCRUM-37: Adaptive hints that reduce over time (3 SP) - Tomasz ✓
- SCRUM-38: Ranking progression system (1 SP) - Kostiantyn ✓
- SCRUM-64: Firebase hosting & React PWA setup (infrastructure) - Kostiantyn ✓

Deliverables: Sprint 4 demo video uploaded to GitHub (Sprint_4_demo.mp4)

What Went Well:

- Excellent team collaboration and communication
- All planned stories completed successfully
- All features delivered on time
- GitHub workflow and testing process improved significantly from previous sprints
- Complete feature set ready for release

What Could Be Improved:

- Sprint extended to two weeks instead of one due to Easter holiday affecting team availability
- Firebase integration issues: compatibility with existing API and unexpected payment requirements caused delays
- Technical risk planning for external services needed improvement

Engineering discipline evidence

Merge pull request #15 from SCRUM-37_Adaptive_hints		Verified	08ce0d5	<>
TomašTouma authored 2 weeks ago · ✓ 3 / 3				
test cases added for SCRUM-37			47f09b2	<>
TomašTouma committed 2 weeks ago · ✓ 6 / 6				
Hint button updated. SCRUM-37			eef5307	<>
TomašTouma committed 2 weeks ago				
reduce available hints by 1 every 3 minutes. SCRUM-37			10317b0	<>
TomašTouma committed 2 weeks ago				
Replace the hardcoded maxHints = 3 and add maxHints as state driven by difficulty			cc8ba81	<>
TomašTouma committed 2 weeks ago				
Commits on Mar 30, 2026				
Sprint 3 demo uploaded		Verified	8379ba2	<>
TomašTouma authored 2 weeks ago · ✓ 3 / 3				
Commits on Mar 26, 2026				
Merge pull request #14 from klostia:SCRUM-36		Verified	061ac05	<>
bogdanabai authored 3 weeks ago · ✓ 3 / 3				
Added back the functionality for the streak as it was accidentally removed			7184b17	<>
bogdanabai committed 3 weeks ago · X 0 / 6				
Updated the ui to show new icon and message if player loses			2f2a14f	<>
bogdanabai committed 3 weeks ago · X 0 / 6				
Added the performance ui			84cad06	<>
bogdanabai committed 3 weeks ago · X 0 / 3				
Added the const for the performance stats			4b359a8	<>
bogdanabai committed 3 weeks ago · X 0 / 3				
Created a function that oathers performance stats on the homepage			~	
Merge pull request #15 from SCRUM-37_Adaptive_hints		Verified	08ce0d5	<>
TomašTouma authored 2 weeks ago · ✓ 3 / 3				
test cases added for SCRUM-37			47f09b2	<>
TomašTouma committed 2 weeks ago · ✓ 6 / 6				
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reduce available hints by 1 every 3 minutes. SCRUM-37			10317b0	<>
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bogdanabai committed 3 weeks ago · X 0 / 3				
Added the const for the performance stats			4b359a8	<>
bogdanabai committed 3 weeks ago · X 0 / 3				
Created a function that oathers performance stats on the homepage			~	

Project Management SB4

SummaryBacklogBoardTimelineList

Filter

SCRUM Sprint 3

19 Mar – 25 Mar (10 work items)

15

56

Complete sprint

SCRUM-31	As a user, I want performance-based rewards so that I stay motivated.	CHALLENGES AND M...	DONE	1	🔴
SCRUM-31	As a player, I want time-based achievements so that I feel rewarded for fast solving.	CHALLENGES AND M...	DONE	1	🔴
SCRUM-25	As a player, I want to see my mistake history so that I can identify patterns and improve.	PERFORMANCE TRAC...	DONE	15	🔵
SCRUM-30	As a learner, I want strategy lessons (e.g., naked pairs, hidden singles) so that I can improve advanced skills.	LEARNING IMPROVE...	DONE	1	🔴
SCRUM-7	As a beginner, I want explanations for hints so that I understand the logic behind them.	LEARNING IMPROVE...	DONE	1	🔴
SCRUM-20	As a beginner, I want guided tutorials so that I can learn basic Sudoku strategies.	LEARNING IMPROVE...	DONE	1	🔵
SCRUM-43	As a player, I want a tutorial replay option so that I can review strategies anytime.	LEARNING IMPROVE...	DONE	1	🔵
SCRUM-15	As a learner, I want strategy lessons so that I can improve advanced skills.	CHALLENGES AND M...	DONE	1	🔵
SCRUM-10	As a beginner, I want beginner-friendly puzzles so that I can build confidence.	SKILL PERSONALIZA...	DONE	1	🔵
SCRUM-36	As a user, I want to see how my current performance compares to past games so that I can measure growth.	PERFORMANCE TRAC...	DONE	1	🔵

SCRUM Sprint 4 26 Mar – 1 Apr (8 work items)		0	10	25	Complete sprint	...
☐ SCRUM-0	As a user, I want statistics on accuracy so that I can track consistency.	PERFORMANCE TRACKING	DONE	5	5	✓
☐ SCRUM-8	As a user, I want a progress dashboard so that I can see long-term improvement trends.	PERFORMANCE TRACKING	DONE	4	4	✓
☐ SCRUM-37	As a player, I want adaptive hints that reduce over time so that I become more independent.	SKILL PERSONALIZATION	DONE	3	3	✓
☐ SCRUM-24	As a user, I want optional smart hints so that I can learn without immediately seeing the answer.	PROGRESS AND SYNC	DONE	3	3	✓
☐ SCRUM-4	As a user, I want push notifications for daily puzzles so that I don't forget to play.	CHALLENGES AND MILESTONES	DONE	3	3	✓
☐ SCRUM-26	As a casual player, I want daily challenges so that I have a reason to return each day.	CHALLENGES AND MILESTONES	DONE	3	3	✓
☐ SCRUM-38	As a user, I want to see my ranking progression so that I feel a sense of achievement.	PERFORMANCE TRACKING	DONE	3	3	✓
☒ SCRUM-64	Firestore hosting, React PWA	DONE	DONE	1	1	✓

Actions

New workflow

All workflows

CI

Management

Caches

Attestations

Runners

Usage metrics

Performance metrics

Showing runs from all workflows

Help us improve GitHub Actions

Tell us how to make GitHub Actions work better for you with three quick questions.

Give feedback

125 workflow runs

Workflow ▾ Event ▾ Status ▾ Branch ▾ Actor ▾

✔ Sprint 4 demo added CI #122: Commit 7ce0373 pushed by TomasToma main	📅 Apr 9 at 11:56 PM GMT+1 🕒 1m 42s ⋮
✔ finalize pwa and ranking features with hosting fix CI #121: Commit 09bb909 pushed by kkostia main	📅 Apr 9 at 8:45 PM GMT+1 🕒 1m 39s ⋮
✔ Merge pull request #20 from kkostia:SCRUM-9 CI #120: Commit 21bbafe pushed by bogdantabalaie main	📅 Apr 9 at 7:46 PM GMT+1 🕒 1m 40s ⋮
✔ SCRUM 9 CI #119: Pull request #20 opened by bogdantabalaie SCRUM-9	📅 Apr 9 at 7:45 PM GMT+1 🕒 39s ⋮
✔ Added the accuracy calculation stats in the display CI #118: Commit 6da550d pushed by bogdantabalaie SCRUM-9	📅 Apr 9 at 7:43 PM GMT+1 🕒 43s ⋮
✔ Added the calculation for the accuracy in the dashboard stats CI #117: Commit 57b348a pushed by bogdantabalaie SCRUM-9	📅 Apr 9 at 7:41 PM GMT+1 🕒 39s ⋮
✔ Added the reset functionality for the mouse CI #116: Commit 57b348a pushed by bogdantabalaie SCRUM-9	📅 Apr 9 at 7:39 PM GMT+1 🕒 39s ⋮

Automated testing & acceptance test evidence

The image displays a development environment with two code editors and a GitHub Actions CI workflow run.

Left Editor (GamePage.jsx): Shows the main game component. It imports `render`, `screen`, `act` from `@testing-library/react`, `userEvent` from `@testing-library/user-event`, `jest-dom` from `@testing-library/jest-dom`, and `GameBoard` from `./Game`. It defines `mockProps` and `baseProps` for testing. A test suite for "Mistake History feature" is shown, including tests for rendering the mistake history section and checking for "No mistakes yet" initially.

Right Editor (Tutorial.test.jsx): Shows the test suite for the `Tutorial` component. It imports `render`, `screen`, `fireEvent` from `@testing-library/react`, `jest-dom` from `@testing-library/jest-dom`, and `Tutorial` from `./Tutorial`. The test suite includes tests for rendering the default strategy, showing the first step by default, moving to the next step when Next is clicked, and moving to the previous step when Prev is clicked.

Bottom Panel (GitHub Actions CI): Shows the CI workflow run for the pull request #20 from `kkostia:SCRUM-9 #120`. The workflow is titled "Merge pull request #20 from kkostia:SCRUM-9 #120" and has a "Re-run all jobs" button. The workflow consists of several jobs, including "build (18.x)", "build (20.x)", "build (22.x)", and "Deploy to Firebase". The "Deploy to Firebase" job is highlighted, showing it succeeded last week in 52s. The job details include a list of steps: "Set up job", "Run actions/checkout@v4", "Use Node.js 20", "Run npm ci", "Run npm run build", "Run FirebaseExtended/action-hosting-deploy@v0", "Post Use Node.js 20", "Post Run actions/checkout@v4", and "Complete job".

Green test results

Actions

New workflow

All workflows

CI

Management

Caches

Attestations

Runners

Usage metrics

Performance metrics

All workflows

Showing runs from all workflows

Filter workflow runs

Help us improve GitHub Actions

125 workflow runs

Sprint 4 demo added

finalize pwa and ranking features with hosting fix

Merge pull request #20 from kkostia:SCRUM-9

SCRUM 9

Added the accuracy calculation stats in the display

Added the calculation for the accuracy in the dashboard stats

Added the reset functionality for the moves

kkostia / SB4_Project

Type to search

CodeIssuesPull requestsActionsProjectsWikiSecurity and qualityInsightsSettings

- Pulse
- Contributors
- Community
- Community standards
- Traffic
- Commits
- Code frequency
- Dependency graph
- Network
- Forks
- Actions usage metrics
- Actions performance metrics

Actions Usage Metrics

Period: Current month

Showing data from 01/04/2026 to 20 minutes ago

Total minutes150

Total job runs150

WorkflowsJobsRuntime OSRunner type

FilterSearch or filter

Workflow	Total minutes	Workflow runs	Jobs	Runner type	Runtime OS
ci.yml	150	48	4	hosted	linux

1-1 of 1Previous1Next

Avg job run time33s

Avg job queue time3s

Job failure rate8%

Failed job usage12

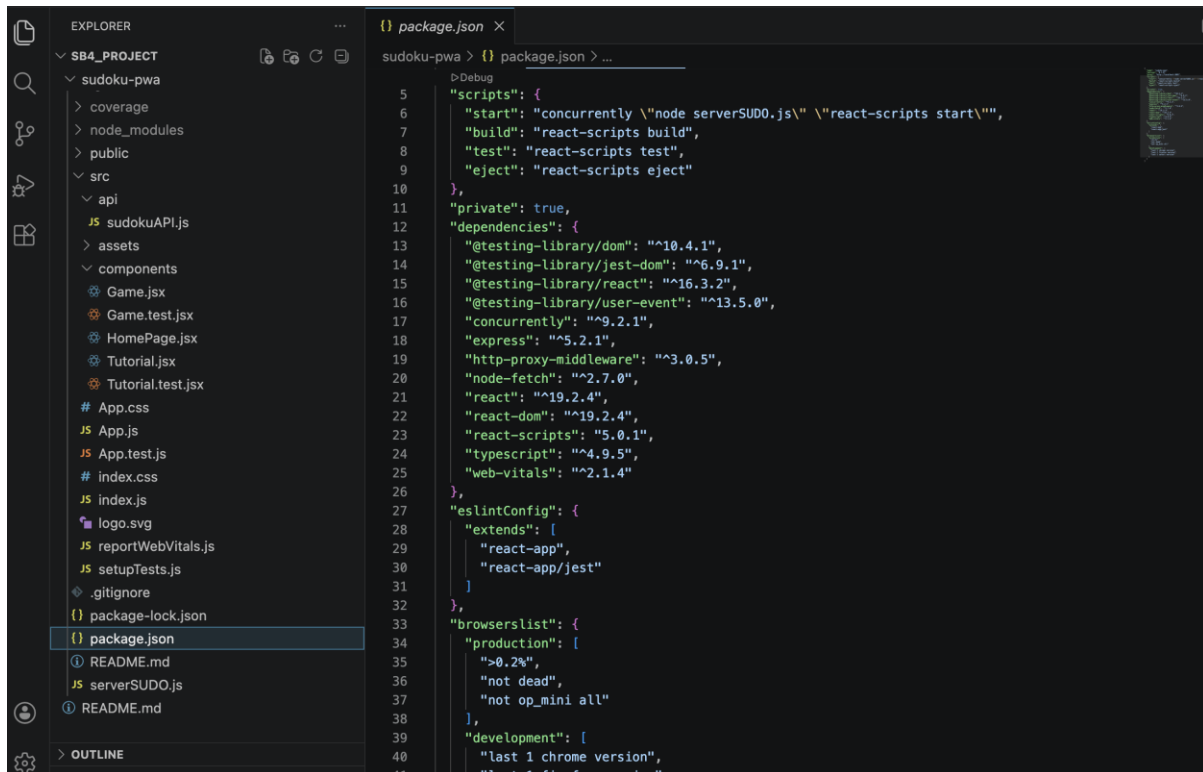
WorkflowsJobsRuntime OSRunner type

FilterSearch or filter

Workflow	Has job failures	Avg run time	Workflow runs	Jobs
ci.yml	25%	43s	48	4

1-1 of 1Previous1Next

Kit Build evidence



```
> sudoku-pwa@0.1.0 build
> react-scripts build output
```

```
(node:29888) [DEP0176] DeprecationWarning: fs.F_OK is deprecated, use fs.constants.F_OK instead
(Use `node --trace-deprecation ...` to show where the warning was created)
Creating an optimized production build...
Compiled successfully.
```

File sizes after gzip:

```
71.97 kB build/static/js/main.280d1d30.js
1.77 kB build/static/js/453.ac6b2737.chunk.js
1.09 kB build/static/css/main.f729efae.css
```

The project was built assuming it is hosted at `/`.
You can control this with the `homepage` field in your `package.json`.

The `build` folder is ready to be deployed.
You may serve it with a static server:

```
npm install -g serve
serve -s build
```

Find out more about deployment here:

<https://cra.link/deployment>

🔍 Search all projects and workspaces

Projects and workspaces ▼



SB4 Sudoku PWA
sb4-sudoku-pwa



Build

Hosting

Downloads (7d total)

2.83MB +179.3%



Deployment history

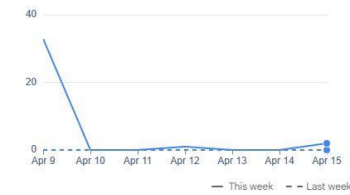
Deployed Apr 9, 2026 11:58 PM

firebase-adminsdk-fbsvc@sb4-sudoku-pwa.iam.gservic...

Functions

Invocations (7d total)

36



Storage

Storage (current)

63.3KB



Your project

Project name SB4 Sudoku PWA ✎

Project ID ⓘ sb4-sudoku-pwa

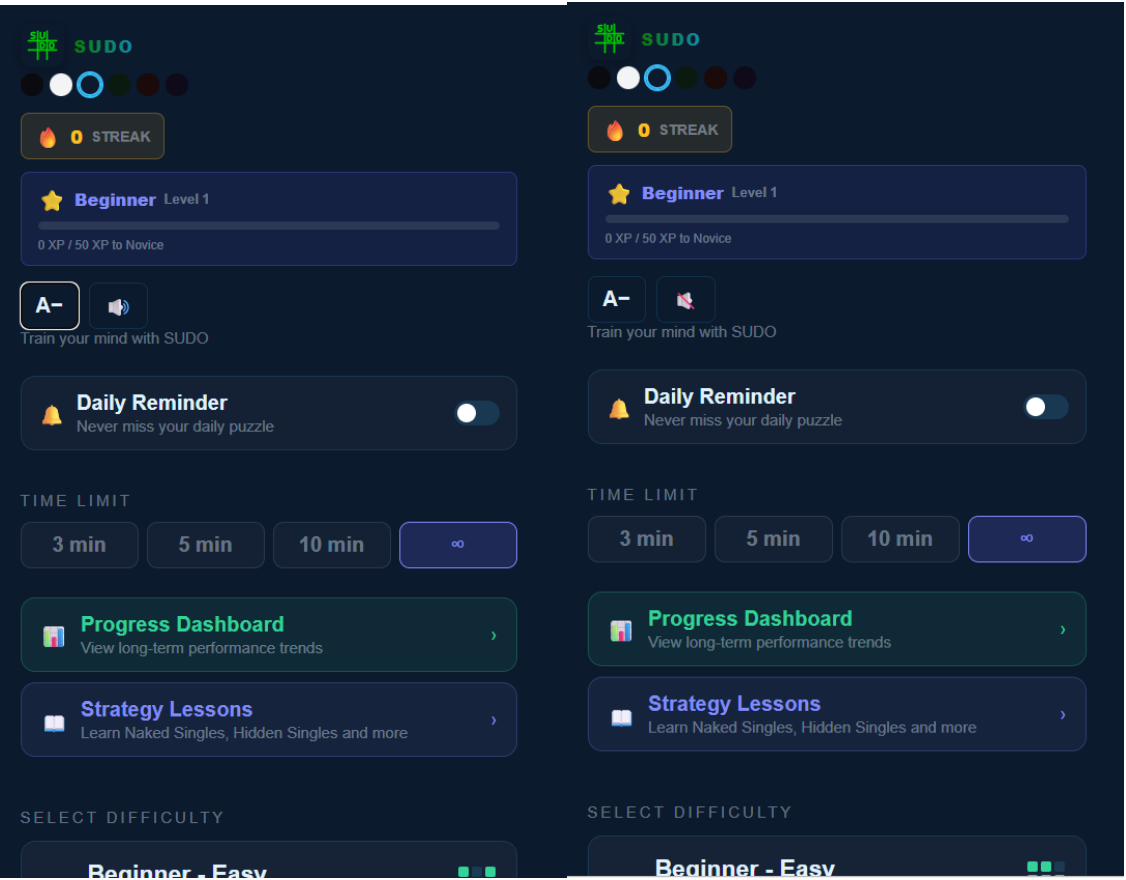
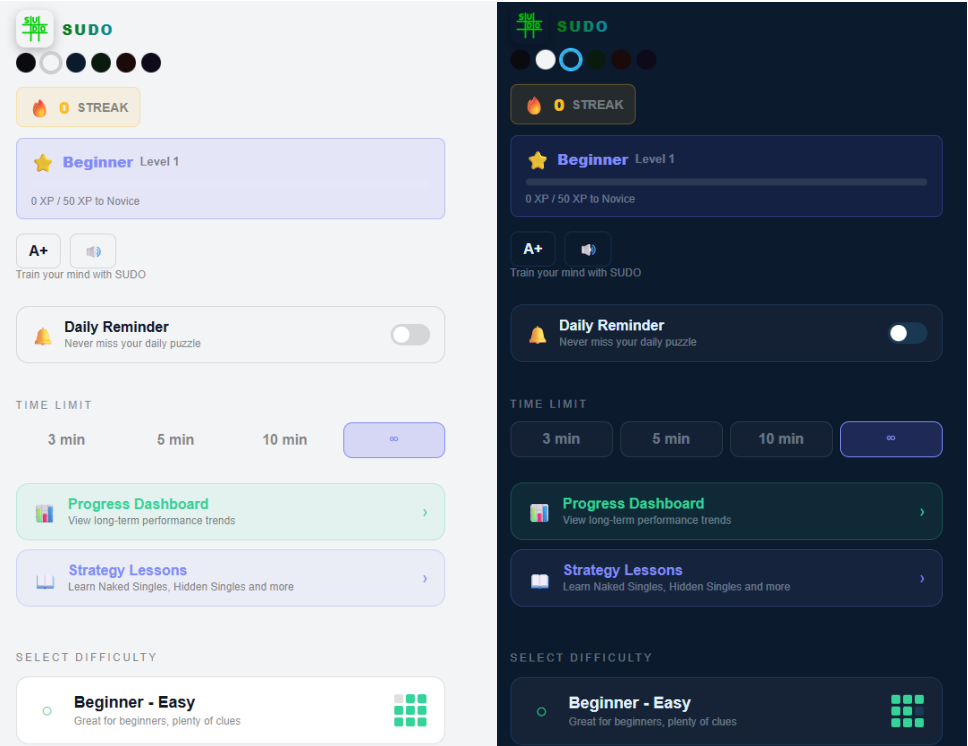
Project number ⓘ 159053624249

Environment

This setting customizes your project for different stages of the app lifecycle

Environment type Unspecified ✎

Screenshots of final application



Progress Dashboard

Total games: 0

Wins: 0

Losses: 0

Win rate: 0%

Current streak: 0

Achievements: 0

Rank: Beginner

XP: 0

STRATEGIES • 1 of 3

① Naked Single

When only one number can go in a cell

① Naked Single

Hidden Single

Box / Line Reduction

A naked single occurs when a cell has only one possible candidate. Every other digit already appears in the same row, column, or 3×3 box.

5	3		7			
6			1	9	5	
	9	8				6
8			6			3
4	2	3	1	?	6	7
7				2		
	6				2	8
			4	1	9	
				8		7

STEP 1/5 Look at the highlighted cell (row 5, col 5).

💡 Pro tip: Scan each empty cell and eliminate the digits already present in its row, column, and box. If only one digit remains, place it!

← Back

Reset

Pause

Intermediate

Mistakes: 0/5

04:57

		8	5				9	
1	2			3	4			
	3	9		7	6			
3	8			9		5		4
			4	1				6
6			2	5	3	7	8	
		4			9		7	5
8		3	1				6	2
9			7		2			3

💡 Get a hint (3 remaining)

Mistake History

No mistakes yet.

← Back

Reset

Pause

Intermediate

Mistakes: 1/5

04:53

		8	5	2			9	
1	2			3	4			
	3	9		7	6			
3	8		2	9		5		4
			4	1				6
6			2	5	3	7	8	
		4			9		7	5
8		3	1				6	2
9			7		2			3

💡 Get a hint (3 remaining)

Mistake History

• Row 4, Col 4: entered 2, correct was 6

1

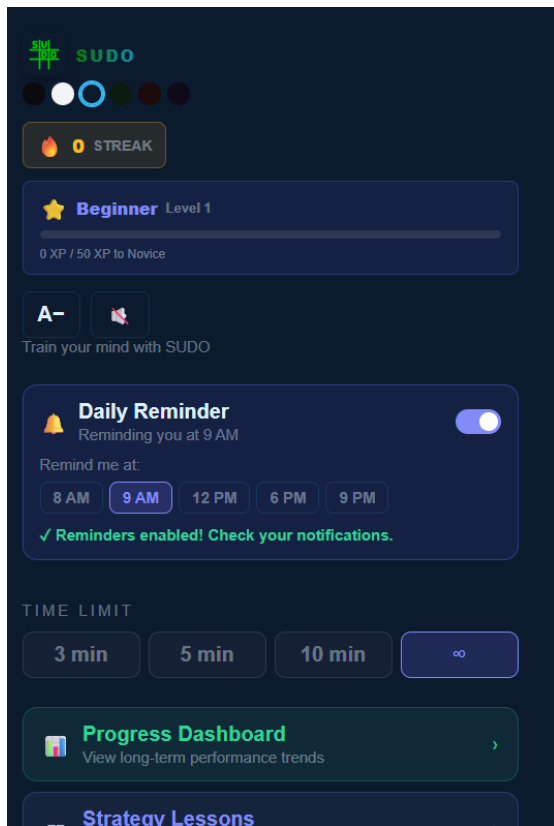
2

3

4

5

6



Individual contributions

Name	ID	Role
Dmytro Senyk	G00439607	Product Owner
Bogdan Tabalae	G00436707	DevOps
Kostiantyn Panasenko	G00436001	Quality Tester
Tomasz Touma	G00439630	Scrum Master

Sprint 1:

SCRUM Sprint 1

5 Mar – 11 Mar (14 work items)

0

0

54

Complete sprint

...

SCRUM-1	As a new user, I want to choose my starting skill level so that I don't feel overwhelmed.	SKILL PERSONALIZA...	DONE	3	=	D
SCRUM-3	As a user, I want to undo moves so that I can fix accidental entries.	CORE GAMEPLAY	DONE	5	^	TT
SCRUM-5	As a player, I want a notes feature so that I can mark possible numbers in cells.	CORE GAMEPLAY	DONE	8	=	D
SCRUM-8	As a player, I want to reset a puzzle so that I can start over if needed.	CORE GAMEPLAY	DONE	3	^	BS
SCRUM-17	As a user, I want to pause a game so that I can resume later.	CORE GAMEPLAY	DONE	4	=	KP
SCRUM-18	As a player, I want visual error highlighting so that I can quickly correct mistakes.	CORE GAMEPLAY	DONE	2	=	BS
SCRUM-32	As a player, I want a clean, simple interface so that I can focus on solving.	CORE GAMEPLAY	DONE	5	^	KP
SCRUM-12	As a player, I want difficulty recommendations so that I choose the right challenge level.	SKILL PERSONALIZA...	DONE	8	=	TT
SCRUM-22	As a user, I want to track my solving time so that I can measure my improvement.	PERFORMANCE TRAC...	DONE	6	^	D
SCRUM-29	As a competitive user, I want timed challenges so that I can test my speed under pressure.	CHALLENGES AND M...	DONE	3	=	TT
SCRUM-34	Remove As a beginner, I want beginner-friendly puzzles so that I can build confidence.		DONE	1	=	D
SCRUM-35	As an advanced player, I want expert-level puzzles so that I stay challenged.	SKILL PERSONALIZA...	DONE	2	=	D
SCRUM-39	As a player, I want mistake limits in challenge mode so that the game feels competitive.	CORE GAMEPLAY	DONE	4	=	BS
☒ SCRUM-63	Create development environment		DONE	0	=	D

Key Assignments:

- **SCRUM-1** (3 SP): Dmytro - As a new user, I want to choose my starting skill level
- **SCRUM-3** (5 SP): Tomasz - Undo moves functionality
- **SCRUM-5** (8 SP): Dmytro - Notes feature for marking possible numbers
- **SCRUM-8** (3 SP): Bogdan - Reset puzzle functionality
- **SCRUM-17** (4 SP): Kosta - Pause game functionality
- **SCRUM-18** (2 SP): Bogdan - Visual error highlighting
- **SCRUM-32** (5 SP): Kosta - Clean, simple interface
- **SCRUM-12** (8 SP): Tomasz - Difficulty recommendations
- **SCRUM-22** (6 SP): Dmytro - Track solving time
- **SCRUM-29** (3 SP): Tomasz - Timed challenges
- **SCRUM-34** (1 SP): Dmytro - Beginner-friendly puzzles
- **SCRUM-35** (2 SP): Dmytro - Expert-level puzzles
- **SCRUM-39** (4 SP): Bogdan - Mistake limits in challenge mode
- **SCRUM-63** (0 SP): Dmytro - Create development environment

Sprint 2:

SCRUM Sprint 2

12 Mar – 18 Mar (6 work items)

4

0

24

Complete sprint

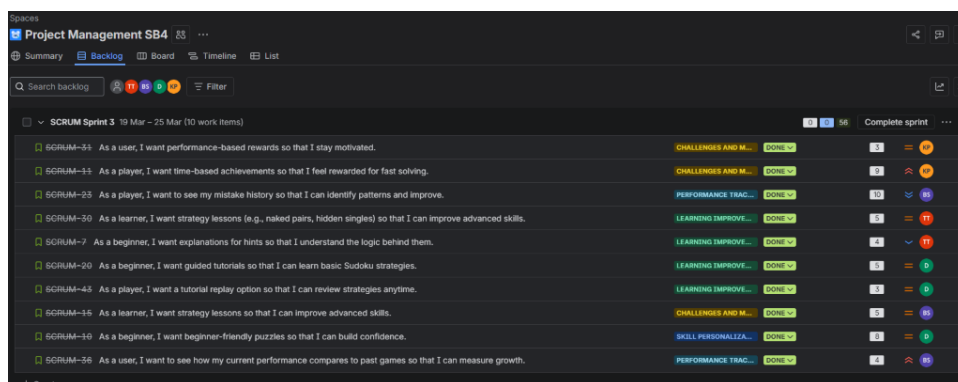
...

SCRUM-13	As a user, I want dark mode so that I can play comfortably at night.	CUSTOMIZATION	DONE	9	^	D
SCRUM-42	As a user, I want accessible font sizes so that the app is easy to read.	CUSTOMIZATION	DONE	4	=	D
SCRUM-14	As a user, I want customizable themes so that I can personalize my experience.	CUSTOMIZATION	DONE	4	^	TT
SCRUM-16	As a player, I want sound effects toggle so that I can control distractions.	CUSTOMIZATION	DONE	2	^	TT
SCRUM-41	As a player, I want a streak counter so that I stay consistent.	PERFORMANCE TRAC...	DONE	5	^	KP
SCRUM-36	As a user, I want to see how my current performance compares to past games so that I can measure gro...	PERFORMANCE TRAC...	TO DO	4	^	BS

Key Assignments:

- **SCRUM-13** (9 SP): D (Dmytro) - Dark mode for night play
- **SCRUM-42** (4 SP): D - Accessible font sizes
- **SCRUM-14** (4 SP): TT (Tomasz) - Customizable themes
- **SCRUM-16** (2 SP): TT - Sound effects toggle
- **SCRUM-41** (5 SP): KP (Kostiantyn) - Streak counter
- **SCRUM-36** (4 SP): BS (Bogdan) - Performance comparison (TO DO - carried to Sprint 3)

Sprint 3:



Key Assignments:

- **SCRUM-31** (3 SP): KP (Kostiantyn) - Performance-based rewards
- **SCRUM-11** (9 SP): KP - Time-based achievements
- **SCRUM-23** (10 SP): BS (Bogdan) - Mistake history tracking
- **SCRUM-30** (5 SP): TT (Tomasz) - Advanced strategy lessons (naked pairs)
- **SCRUM-7** (4 SP): TT - Explanations for hints
- **SCRUM-20** (5 SP): D (Dmytro) - Guided tutorials
- **SCRUM-43** (3 SP): D - Tutorial replay option
- **SCRUM-15** (5 SP): BS - Strategy lessons
- **SCRUM-10** (8 SP): D - Beginner-friendly puzzles
- **SCRUM-36** (4 SP): BS - Performance comparison (completed from Sprint 2)

Sprint 4:

SCRUM Sprint 4 26 Mar – 1 Apr (9 work items)			13	1	25	Complete sprint	...
SCRUM-9	As a user, I want statistics on accuracy so that I can track consistency.	PERFORMANCE TRACKING	DONE	13	1	25	
SCRUM-6	As a user, I want a progress dashboard so that I can see long-term improvement trends.	PERFORMANCE TRACKING	DONE	13	1	25	
SCRUM-37	As a player, I want adaptive hints that reduce over time so that I become more independent.	SKILL PERSONALIZATION	DONE	13	1	25	
SCRUM-24	As a user, I want optional smart hints so that I can learn without immediately seeing the answer.	PROGRESS AND SYNC	DONE	13	1	25	
SCRUM-4	As a user, I want push notifications for daily puzzles so that I don't forget to play.	CHALLENGES AND MILESTONES	DONE	13	1	25	
SCRUM-26	As a casual player, I want daily challenges so that I have a reason to return each day.	CHALLENGES AND MILESTONES	DONE	13	1	25	
SCRUM-38	As a user, I want to see my ranking progression so that I feel a sense of achievement.	PERFORMANCE TRACKING	DONE	13	1	25	
SCRUM-64	Firestore hosting, React PWA		DONE				

Key Assignments:

- **SCRUM-9** (6 SP): BS (Bogdan) - Statistics on accuracy
- **SCRUM-6** (4 SP): BS - Progress dashboard
- **SCRUM-37** (3 SP): TT (Tomasz) - Adaptive hints that reduce over time
- **SCRUM-24** (5 SP): TT - Optional smart hints
- **SCRUM-4** (5 SP): D (Dmytro) - Push notifications for daily puzzles
- **SCRUM-26** (1 SP): D - Daily challenges
- **SCRUM-38** (1 SP): KP (Kostiantyn) - Ranking progression system
- **SCRUM-64** (0 SP): KP - Firebase hosting & React PWA setup

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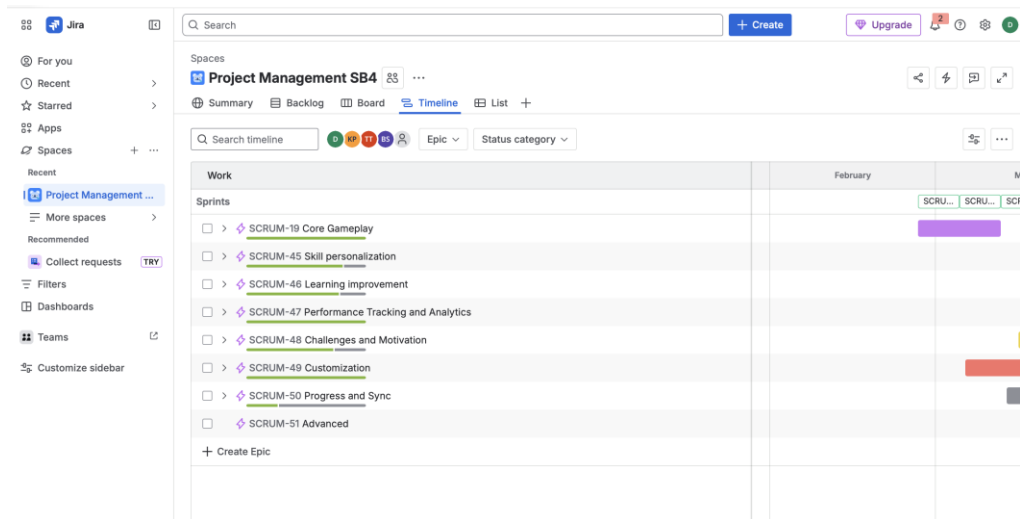
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Continuous Integration & Deployment

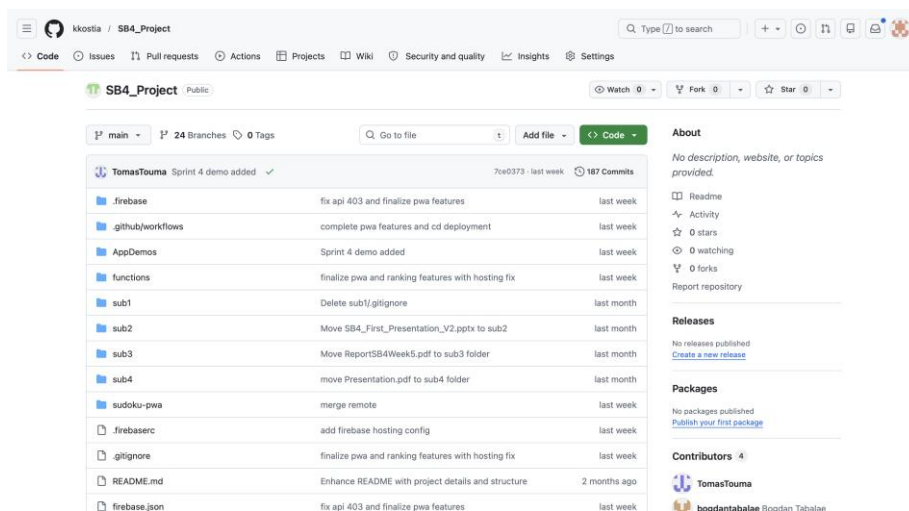
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Required Links (In the PDF)

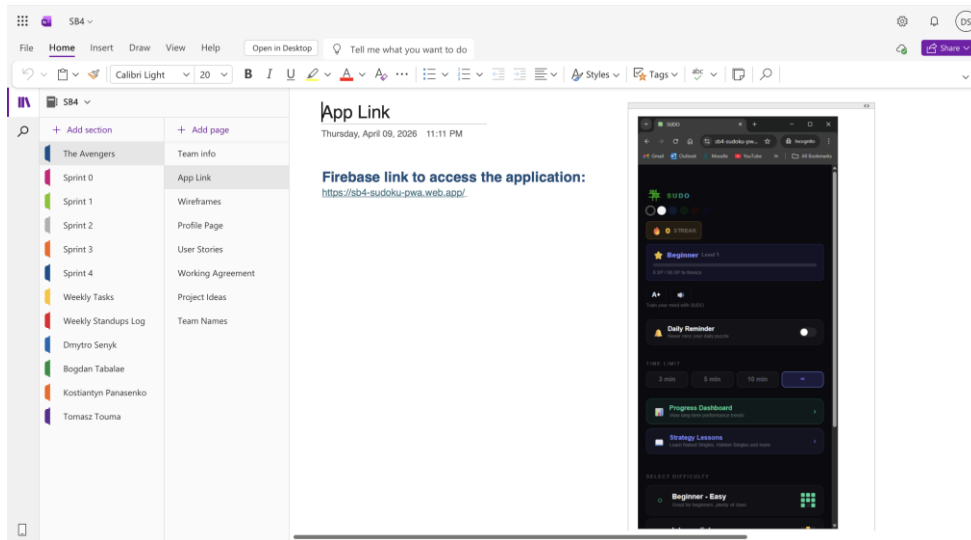
Jira: <http://atu-team-g7ggluv.atlassian.net/>



GitHub: https://github.com/kkostia/SB4_Project



OneNote: https://atlantictu-my.sharepoint.com/:o:/r/personal/g00439607_atu_ie/Documents/SB4?d=wa0d90890bde740d4a2ee5e052aed8244&csf=1&web=1&e=eKEnJX



MSProject: https://atlantictu-my.sharepoint.com/:u:/g/personal/g00439607_atu_ie/IQAOacfsKxmRRbvJVI8kzc3YAWd1qKM7GtNTz3gyfsnl540?e=bkXdM3

